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| <b>4a</b>                 | <p>Using</p> $R = \frac{\rho l}{A}$ $= \frac{7.9 \times 10^{-7} (l)}{\pi (6.0 \times 10^{-6})^2}$ $l = 0.137 \text{ m}$                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>4bi</b>                | The resistance at a point on the curve can be determined using the ratio of voltage to the current at that point.                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>4bii</b>               | <p>As <math>V</math> increases, temperature increases, the no. of charge carriers (positive holes and/or negative electrons) per unit volume increases.</p> <p>At the same time, the amplitude of vibration of atomic cores also increases, resulting in more frequent collisions with the atomic cores.</p> <p>The increase in conductivity due to more charge carriers is more significant than the decrease in conductivity due to increased thermal vibration.</p> <p>Hence resistance of thermistor decreases.</p> |
| <b>4biii</b><br><b>1.</b> | $R_{xy} = \left( \frac{1}{960} + \frac{1}{1250} \right)^{-1} = 543 \Omega$                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>4bii</b><br><b>2.</b>  | <p><i>By Potential Divider Principle,</i></p> $V_{xy} = \frac{R_{xy}}{R_{xy} + 7800} \times 1.5$ $= \frac{543}{543 + 7800} \times 1.5$ $= 0.0976 \text{ V}$                                                                                                                                                                                                                                                                                                                                                             |
| <b>4bii</b><br><b>3.</b>  | <p>When temperature drops, resistance of thermistor increases.</p> <p>Hence, the effective resistance of XY increases.</p> <p>By Potential Divider Principle, the potential difference across XY increases.</p>                                                                                                                                                                                                                                                                                                         |

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|  | Since power is proportional to $V^2$ for constant lamp resistance, the power dissipated in the filament lamp increases. Thus, the bulb is brighter. |
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